Stanford OHS

Mitigating a Global Health Crisis: Proposals Towards Improving Automated External Defibrillator Accessibility

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Abstract

Sudden Cardiac Arrest is a major global public health concern that accounts for 15-20% of worldwide deaths[63]. There are more than 356,000 out-of-hospital cardiac arrests in the United States each year, with around a 10% survival rate[12]. Current endeavors to address this issue center predominantly around the development of innovative medical devices with the goal of engineering a permanent solution. However, the pace of progress in this enterprise is increasingly proving inadequate in yielding promising outcomes in the near future. A prioritized focus should instead remain on utilizing current technologies, such as the Automated External Defibrillator (AED), effectively. Specifically, developing strategies that facilitate faster access to these devices when they are a critical necessity may offer a practical course toward significantly improving the survival rate of Sudden Cardiac Arrest in the short term. A significant obstacle to the accessibility of AEDs is their unaffordability for lower-to-middle-income families. Furthermore, AEDs are commonly located in public areas such as shopping malls or schools, whereas many deaths related to cardiovascular diseases occur in private locations that lack easily- accessible AEDs. In this context, I propose a set of innovative measures aimed at improving AED accessibility. The primary proposal involves the establishment of "Medical Emergency Boxes" (MEBs), which would bring AEDs to communities and rural areas as an equally accessible public good. Methods that promote AED accessibility, such as the MEB system, AED-locating applications, or the utilization of existing software in wearable devices like Apple Watches, may offer a short-term solution to this pressing global health crisis.

Keywords: Automated External Defibrillator; Sudden Cardiac Arrest; Cardiovascular System

Introduction

Valenzuela TD et al. reported that the chances of survival following cardiac arrest decrease by 10-15% for every minute that cardiopulmonary resuscitation is delayed[76]. The Automated External Defibrillator (AED) is a contemporary medical device that non-medical professionals can use and has a 90% success rate in preserving cognitive function if defibrillation is applied within ten minutes of victim collapse[76]. This makes AEDs efficient and effective in restoring executive function following Sudden Cardiac Arrest. Despite the widespread availability of AEDs in public areas such as schools, stadiums, shopping malls, and airports, the global fatality rate from Sudden Cardiac Arrest remains significant, with 7 out of 10 incidents occurring in private locations, such as the victim's home[12]. The main complication remains the inability to efficiently access an AED. Previous studies have shown that various factors, such as race, sex, and income (which are linked to distance from critical care hospitals), affect Sudden Cardiac Arrest survival rates. This paper emphasizes the importance of considering medical inequalities in the context of Sudden Cardiac Arrest and defibrillation when developing possible strategies to improve AED accessibility.

Review of the Cardiovascular System

The Cardiovascular System is a major organ system that encompasses the heart and a network of blood vessels responsible for the circulation and transport of blood and various materials throughout the human body. The human circulatory system is referred to as a double circulatory system due to the existence of two separate circuits of blood flow. The Pulmonary Circuit is that in which the right side of the heart transports oxygen-poor blood towards gas-exchange tissues, whereas the second circuit, the Systemic Circuit, is characterized by the left side of the heart pumping oxygenated blood to organs and tissues of the body and returning oxygen-poor blood to the heart. This process is essential for providing oxygen and nutrients to the body's cells and removing metabolic waste products[36].

Blood Vessels

There are three primary types of blood vessels: arteries, capillaries, and veins. Arteries, muscular tubes lined with smooth tissues consisting of three layers–Intima, Media, and Adventitia–deliver oxygen-rich blood from the heart to organs, with the exception of the pulmonary artery, which delivers oxygen-poor blood to the lungs[13]. The Intima is the innermost layer lined by endothelium tissue and ensures the smooth, one-directional flow of blood; the Media, found in between the Intima and Adventitia, is a layer of muscle that enables arteries to cope with

the high pressures that come from the heart; the Adventitia is a layer of connective tissue that anchors arteries to neighboring tissues. With organ tissues, arteries branch into arterioles, smaller vessels that transmit blood to capillaries. Capillaries, the microscopic vessels found within porous, single-layer tissue, exchange gasses, proteins, fats, and other chemicals via diffusion or active pressure. Capillaries tend to function in an interwoven network referred to as a capillary bed; cells that are more metabolically active require an increased amount of capillaries to adequately supply the nutrients required. Within a capillary bed are two vessel types– the true capillaries that function in circulation, and vascular shunts, which directly connect arterioles and venules on the polar ends of the capillary bed, permitting for a bypass mechanism[13]. Veins are similar to arteries in structure yet are differentiated via the direction of blood flow. Arteries transport blood away from the heart, whereas veins convey blood toward the heart. The exception is the Portal Veins, which act as a pathway of blood flow between capillary bed pairs.

Structure of the Heart

In the Cardiovascular system, the heart drives circulation via metabolic energy, elevating blood hydrostatic pressure– the pressure that blood exerts on the vessels it surrounds[36]. AEDs directly affect the functioning of the heart, and so an understanding of the heart's mechanisms, from both a chemical and organizational scale, is helpful in understanding how AEDs work. The heart, a slightly-coned shaped organ located in the mediastinum of the thoracic cavity, rests above the diaphragm and behind the sternum, between the lungs[46]. The four chambers of the heart are morphologically and functionally distinct from one another. Oxygen-poor blood from the right atrium and ventricle is moved into the lungs; oxygenated blood enters the left atrium, and the ventricle pumps blood around the body. These chambers are separated by industrial and interventricular septa, and the atrioventricular and ventricular-arterial valves maintain the unidirectional flow of blood[46,36]. Heart wall layers are composed of three layers– the epicardium, which covers the surface of the heart (also referred to as the visceral pericardium), the myocardium, the muscular middle layer, and the endocardium, the thin, smooth innermost membrane **(Figure 1)**. The myocardium consists of two cell types: the cardiac muscle cells, which are striated, branching cells containing many mitochondria to power synchronous contraction, and the cardiac myocytes, the striated fibrous cardiac skeleton cells[46,36].

Cellular Composition and Mechanisms

The myocardium consists of two cell types: the cardiac muscle cells, which are striated, branching cells containing many mitochondria to power synchronous contraction, and the cardiac myocytes, the striated fibrous cardiac skeleton cells[46,36]. Cardiac fibroblasts function to maintain the structural integrity of the heart; the cells can undergo transdifferentiation (the process by which the cell turns into another cell type) into myofibroblasts. With around three billion cardiac myocyte cells in the heart, these cells comprise around 75% of the total volume of the myocardium[46,82.] A myocyte is composed of bundles of myofibrils that consist of myofilaments. A key characteristic of myofibrils is the sarcomere– the repeating, fundamental units of contraction within the cell that are organized by myosin and actin filaments **(Figure 2)**. When chemical and mechanical interactions occur between the two proteins, the sarcomere length shortens, causing the myocyte cell to contract. Changes in sar-

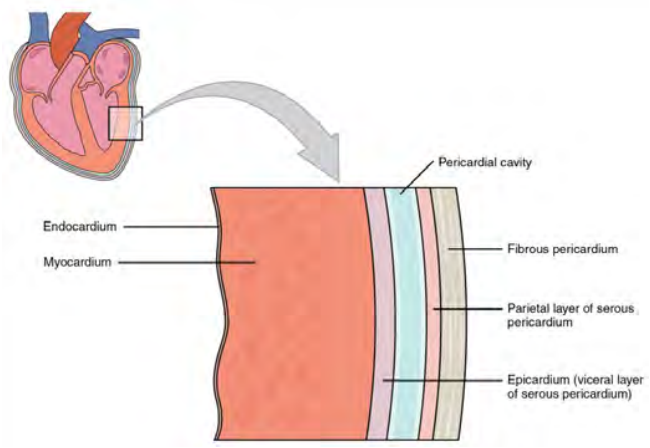


Figure 1 The three heart layers– the epicardium, which covers the surface of the heart (also referred to as the visceral pericardium), the myocardium, the muscular middle layer, and the endocardium, the thin, smooth innermost membrane. The pericardial cavity is a space containing serous fluid that acts as lubrication to facilitate heart movement. The Fibrous pericardium is not considered a heart layer but rather is a connective tissue that provides support. The parietal layer is the outermost layer of the serous pericardium[34].

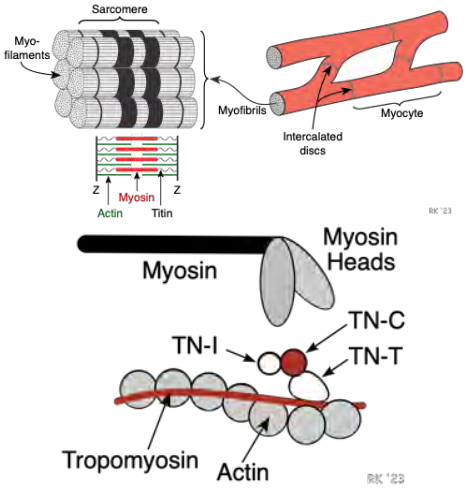


Figure 2 A. Sarcomeres are composed of microfilaments – myosin and actin. Changes in sarcomere length are the mechanism by which the heart regulates its force of contraction. **B.** Expanded depiction of "ratcheting. " The two myosin heads interact with the binding site on the regulatory protein complex composed of actin, tropomyosin, and troponin. The binding at the actin-myosin complex induces ATP hydrolysis, the catabolic reaction that releases energy for the conformational change, leading to a sliding movement of the myosin heads and the actin[21,22,24].

comere length are a foundational mechanism by which the heart regulates its force of contraction **(Figure 2)**. When the sarcomere length increases, protein binding-site troponin-C (TN-C) increases in affinity for Calcium ions (four calcium bind to one troponin-C site), causing increased tension development in the myocyte, increasing contraction of the heart[21] **(Figure 2)**. In short, myocyte cells maintain and direct the mechanical and electrical functions of the heart– specifically responsible for the cardiac contraction-relaxation cycle[21]. During a cardiac cycle, ions are transported across the cardiomyocyte cell membrane, changing the membrane voltage (Vm). Calcium ions entering the myocytes prompt the contraction of the cardiac muscle of the atria and, following a short pause at the atrioventricular node- the tissue that conducts electrical signals from the atria to the ventricles–ventricles contract[56,57]. This electrochemical change within the myocardium marks the initiation of the cardiac cycle, a series of pressure changes that leads to blood movement around the chambers of the heart and body.

The Cardiac Cycle

Diastole and Systole

There are two periods of the cardiac cycle– diastole and systole (**Figure 3, 4**). Diastole refers to the period in which the ventricles are relaxed; during this phase, blood flow is passive from the left atrium and right atrium into the left ventricle and right ventricle, respectively[44]. The blood flows through the mitral and tricuspid valves– the atrioventricular valves– that are located between the atria and the ventricles (**Figure 3, 4**). The right atrium receives venous, oxygen-poor blood from the body via the superior vena cava and the inferior vena cava (**Figure 3, 4**). The left atrium receives oxygen-rich blood from the lungs via four pulmonary veins. The atria contract, propelling blood into the ventricles at the end of diastole (**Figure 3, 4**). Systole refers to the period in which the ventricles contract, ejecting blood through the aortic and pulmonic valves and into the aorta and pulmonary artery[44]. Here the atrioventricular valves are shut, and as such, there is no blood flow entering the ventricles; rather, blood enters the atria through the vena cavae and pulmonary veins[44,21,22](**Figure 3, 4**).

The Seven Phases

The cardiac cycle is then divided into seven key phases[21]. The first phase is that of atrial contraction, represented by the P-wave on an electrocardiogram. Having received specific signals from the sinoatrial node–the cluster of muscle fibers located in the right atrium– cardiac myocytes depolarize[56]. As the atria contract, atrial chamber pressure increases, and this causes the blood flow through the atrioventricular valves; here, the diastolic blood pressure (DP) in an artery is at its lowest pressure as the ventricles relax and fill[21,22]. Ventricular depolarization is marked by the QRS wave on the electrocardiogram and is generally referred to as the Isovolumetric phase of the cardiac cycle. The depolarization triggers excitation-contraction coupling–the cellular process that leads to myocyte contraction[21,22,24]. As action potentials travel through the sarcolemma, down into the transverse tubule system (extensions of the cell membrane), dihydropyridine receptors of L-Type calcium channels open, allowing calcium entry into the cell. The influx of calcium ions triggers theryanodine receptors to release calcium from the sarcoplasmic reticulum– a region of the smooth endoplasmic reticulum of cardiac muscle– increasing intracellular calcium

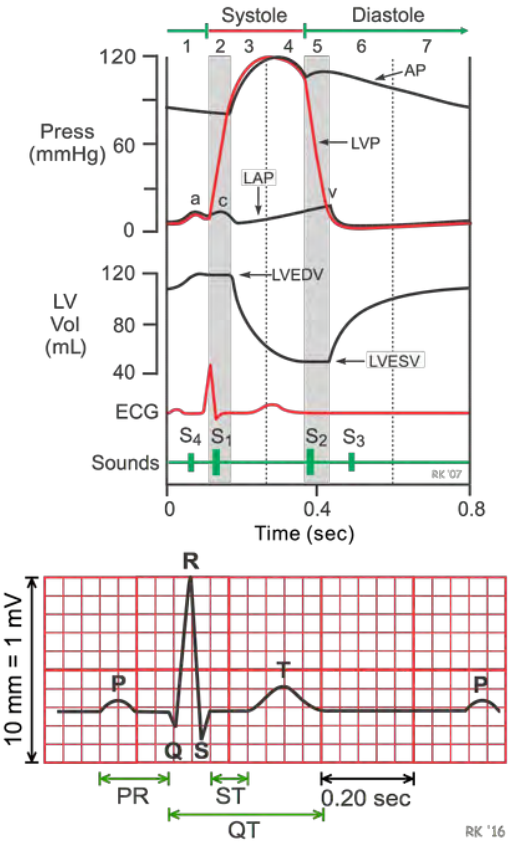


Figure 3 The cardiac cycle diagram illustrates the aortic pressure (AP) changes, the left atrial pressure (LAP), left ventricular pressure (LVP), and left ventricular volume (LV Vol), during one cycle of contraction-relaxation in relation to time to the electrocardiogram. The systole period takes place when the ventricles contract and diastole refers to the period where the ventricles relax. The electrocardiogram is expanded below. Atrial contraction is represented by the P-wave. Ventricular depolarization is marked by the QRS wave. The T-wave represents ventricular re-polarization[21,22].

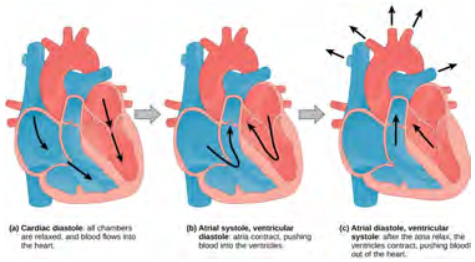


Figure 4 The cardiac cycle is a complex series of events that occur during a single heartbeat, involving both the contraction and relaxation of the heart muscles. This cycle can be divided into two main phases: systole, where the heart muscles contract, and diastole, where the heart muscles relax. During systole, the ventricles contract and force blood out of the heart and into the pulmonary artery and aorta, while the atria are in a relaxed state and filling with blood. During diastole, the ventricles relax and fill up with blood from the atria. The atria then contract, helping to push the remaining blood into the ventricles.[21,22]

concentration[21,22,24]. The excess calcium binds to Troponin-C, the aforementioned binding site of calcium ions. Following this binding process, a conformational change is induced such that Troponin-I– another protein regulatory complex–permits the binding of an actin molecule to the myosin ATPase located on the myosin head. The binding at the actin-myosin complex induces ATP hydrolysis, the catabolic reaction that releases energy for the conformational change, leading to a sliding movement of the myosin heads and the actin, referred to as "ratcheting" **(Figure 2)**. These ratcheting cycles shorten the sarcomere length, and it occurs for as long as there is elevated calcium[21,22,24]. The myocyte contraction leads to a rapid increase in intraventricular pressure (maximal dP/dt). With this pressure, the atrioventricular valves close, resulting in the initial heart sound (S_1)[22,]. At the end of this phase, the calcium entry into the cell gradually slows, and the calcium is eventually removed from Troponin-C via the sacro-endoplasmic reticulum calcium-ATPase pump, which lowers the cytosolic calcium concentration. When the calcium unbinds from the Troponin-C complex, another conformational change occurs, causing, again, inhibition at the Troponin-I actin-binding site, resulting in the initial sarcomere being restored to its original length[24]. The opening of the aortic valve characterizes the third phase of the cardiac cycle. Here, blood is discharged when the intraventricular pressure rises, exceeding the pressures of the aorta and pulmonary artery, causing the respective aortic and pulmonic valves to open. With the ejection of blood, the energy gradient pushes blood from the ventricles into the aorta and pulmonary artery. The following stage is the "ejection" phase, where the T-wave is visible on the electrocardiogram representing ventricular repolarization. The repolarization causes a decreased ventricular tension and a decreased rate of blood flow out of the left ventricle; simultaneously, the pressures of the left and right atrium rise as venous blood returns from the lungs. The second heart sound (S_2). occurs when the aortic and the pulmonic valves close, which marks isovolumetric relaxation. A small backflow of blood into the ventricles and an incisura, or dicrotic notch, in the pulmonary and aortic artery, marks the end of systole and the being of diastole[21]. In this phase, the volume of the blood that remains in the ventricle is the end-systolic volume, and the difference between the end-diastolic volume and end-systolic volume is referred to as the stroke volume. The left atrial pressure rises as oxygenated blood returns from the lungs, and the peak pressure of this stage is referred to as the v-wave on the electrocardiogram. Following the closure of the aorta, the isovolumetric pressures drop, falling below their respective atrial pressures. During this period, the atrioventricular valves open, and passive blood flow into the ventricles commences; ventricles are still relaxing[21]. The final phase of the cardiac cycle is referred to as the "filling phase," as the ventricles expand, causing rising intraventricular pressures. The bi-leaflet mitral valve–which connects the left atrium and the left ventricle– opens due to pressure changes, permitting blood to flow passively into the left ventricle. In a functioning heart, the ventricles are 90% full prior to atrial contraction (phase 1)[22]

Arrhythmias

Disruptions in the cardiac cycle can lead to arrhythmias–an irregular heartbeat. Cardiac arrhythmias, of which there are three primary types and two subtypes, are categorized by the speed at which the heart is beating and the point of origin. Tachycardia refers to an abnormally fast heart rate, often over 100 bpm[68]. Bradycardia refers to cases in which the heart rate falls significantly slower than normal, often falling below 60 bpm[8,69]. Premature Heartbeats, where an extra heartbeat is detected, are common arrhythmias and tend to be harmless[4,]. Supraventricular arrhythmias and Ventricular arrhythmias are tachycardia[4,24,63]. Supraventricular arrhythmias occur specifically in the atrioventricular node. Ventricular arrhythmias originate in the lower chambers of the heart and can lead to sudden death. Cardiac arrhythmias are the leading cause of cardiac arrest, and as such, arrhythmias are one of the primary preductors that the use of an AED machine may soon be necessary[63].

Heart Conditions

Heart failure, Sudden Cardiac Arrest, and Heart Attacks are separate conditions. In brief overview, heart failure occurs when the heart is unable to eject enough blood into the body due to an electrical malfunction, and this differs from a heart attack–when the coronary artery is either blocked or narrowed excessively, preventing blood flow to a specific region of the heart itself. It is important to note that heart failure and heart attack do not, by definition, mean the heart has stopped beating, though both heart failure and heart attack can lead to Sudden Cardiac Arrest– the condition in which there is no longer any blood flow as the heart has completely ceased activity. Almost 90% of cardiac arrests outside of hospitals are fatal. In one year, around 440,000 Americans die from Cardiac Arrest, with 350,000 of these cases occurring out-of-hospital (OHCA)[12,74,75]. As such, cardiac arrest is a public health issue, and for this paper, a review of Sudden Cardiac Arrest is important to understand in which situations AED usage is necessary. The primary causes of cardiac arrest in adults include cardiac arrhythmia, shock, and electrolyte imbalances. The most common cause is Cardiac Arrhythmia, such as Ventricular Fibrillation, Ventricular Tachycardia, and Bradycardia[4,63]. Shock is a medical condition with various underlying causes and is defined as a state in which blood pressure is significantly low, ceasing blood flow to the brain, often causing the patient to lose consciousness[37]. Finally, when electrolytes, such as potassium and calcium, are not in balance within the cell membrane, the heart is unable to respond to the electrical impulses properly. Without the internal chemical gradient functioning, the heart fails to beat, disrupting the cardiac cycle. In minors, causes of Sudden Cardiac Arrest often occur due to structural abnormalities– such as Congenital Heart Disease or Hypertrophic Cardiomyopathy–; Myocarditis, an inflammation of the heart muscle resulting from viral or bacterial infections, certain allergens or toxins that affect the heart's ability to contract, and genetic conditions[48]. Both healthy adults and children can also experience Sudden Cardiac Arrest from Commotio Cordis, in which a strike or blunt force to the chest disrupts the heart's electrical system, causing arrhythmia and death[3].

Socioeconomic and Racial Disparities

According to the American Heart Association, men, specifically black athletes, have a statistically higher risk of Sudden Cardiac Arrest[6,16,75]. Further, low-income African American neighborhoods and predominantly Hispanic/Latino neighborhoods face lower rates of bystander Cardiopulmonary Resuscitation (CPR) compared to higher-income, predominantly White neighborhoods[16]. At the center of health inequality is ethnicity, which tends to correlate with income levels, individual access

to rapid medical assistance, and to a degree, genetic differences and diet[84]. As aforementioned, cardiovascular disease is the primary cause of cardiac arrest, and specific ethnicities encounter a higher risk for diabetes, hypertension, and, in general, heart disease [81]. In the United States, African Americans are at a statistically higher risk for hypertension, with African American and Mexican American adults having among the highest rates of hypertension globally[31, 84]. Hypertension increases the risk of Sudden Cardiac Arrest significantly–African Americans have a higher incidence of out-of-hospital cardiac arrest (and, in some studies, face lower survival rates) compared to whites[31]. Researchers Di Zhao et al. found that the "single most important factor" to explain this disparity was income (50.5%), followed by hypertension (22.1%), diabetes mellitus (19.6%), and education (19.1%)[84]. In fact, these risk factors explained 65% of the race difference in Sudden Cardiac Arrest, and the racial disparities observed illustrate a major public health concern[84]. African American persons and other minorities have less access to medical care in the United States, which has been found to correlate with their higher rates of unemployment and underrepresentation in jobs that include health insurance[7]. Lack of access to medical care – specifically preventive care, early intervention, etc.– can account, in part, for the different rates of Sudden Cardiac Arrest between racial groups. This accounts for higher-income areas being predominantly white, whereas lower-income areas are comprised of African Americans, Hispanics, and other minorities **(Figure 5)**. Proximity to hospitals and ICUs differs between higher and lower-income communities, with higher-income areas with higher access to healthcare services[26]. Eberth et al. conducted a study to confirm that travel times to hospital services were longer for rural, low-income areas with high minority representation, specifically African American, in the United States[26] **(Figure 5)**. In other words, there is there a correlation between African American/minority communities and access to a hospital or clinic with an AED; specifically, minority cluster communities tend to be situated in areas that have significant emergency service accessibility barriers compared to predominantly white residential areas[26,36]. Thus, in general, individuals from minority groups face more significant AED accessibility barriers compared to higher-income, white individuals[36]. Further, rural, low-income areas also tend to have lower education levels– lower education levels correlate with less knowledge about cardiac disease/failure symptoms and less familiarity with how to perform CPR[16,26,36]. The lack of easy access to immediate health care services– and thus devices, such as AED machines– also plays a considerable role in the lower survival rates for Cardiac Arrest in these communities[36].

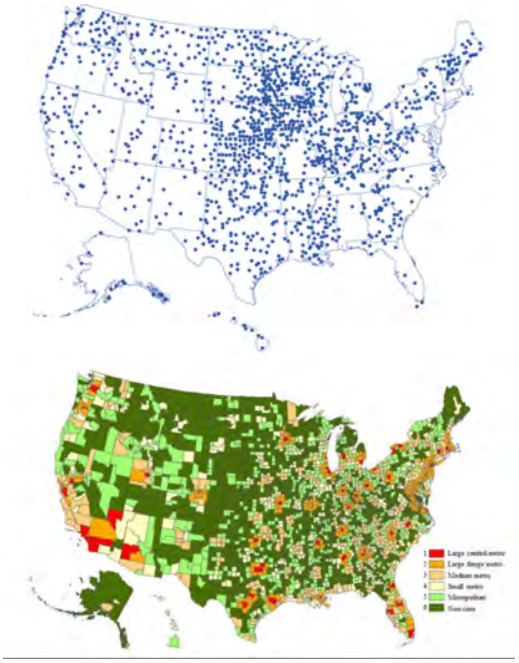


Figure 5 . **A.** Distribution of critical access hospitals in the United States, 2021. Hospitals are represented by blue dots. **B.** Map distribution of rural and metro areas in the United States. Note how there exists a correlation between "non-core" (or rural) areas and a significantly lower number of critical access hospitals[36, 86]

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Cardiopulmonary Resuscitation

Cardiopulmonary Resuscitation (CPR) is an increasingly well-known medical practice that any individual, regardless of ethnicity, language, or gender, can perform. According to Zelfani et al. CPR is the primary method to "buy time" when waiting for a defibrillator or professional medical staff, and immediate bystander CPR can increase an individual chance of survival two-to-fourfold[83]. Ventricular fibrillation and ventricular tachycardia without a pulse tend to be most commonly observed following cardiac arrest, and CPR can prolong such rhythms, thus increasing the chances of successful defibrillation[83]. For maximum survival, CPR should be performed less than four minutes following the collapse[30]. After four minutes, cerebral deterioration often occurs, which not only lowers survival chances but is likely to lead to long-term cognitive impairments (such as memory or attention difficulties, language complications) or executive function deficits in cases where the patient is successfully released from the hospital[30]. Responders are trained to perform CPR via a 30:2 compression-to-ventilation ratio, as each compression pumps blood to the brain and organs mechanically [17]. Around 100 to 120 beats per minute are the recommended rate, and often pressing down on the sternum, located between the nipples, to the rhythm of "Stayin' Alive" by the Bees Gees is a practical method to keep count of time under pressure[30,17]. Following the 30 compressions, pinch the nose of the victim tightly to prevent airflow. Place your mouth directly over the victims, and breathe heavily twice. If there is one on hand, place a CPR disease-prevention mask on top of the victim's lips prior to the two breaths. Immediately continue compression afterward. Note that CPR of adults and infants differ slightly in CPR protocol, in which two fingers rather than the palm of the hand are required to perform the compression. CPR success, however, can depend on a variety of factors. In cases of Sudden Cardiac Arrest in public, 45% of men received CPR assistance compared to only 39% of women, meaning men are 1.23 more likely to receive help, and thus 23% more likely to survive[17,50]. When surveyed, bystanders responded that differences in the male and female anatomy contributed to the hesitation; women having breasts makes some individuals uncomfortable as to whether or not they should intervene[50]. The American Heart Association notes that only 46% of victims get CPR from bystanders[74,75,]. Currently, 9 out of 10 victims of cardiac arrest die; CPR, however, can double or triple the chances of one's survival[16]. Yet around 70% of Americans are hesitant to perform CPR due to a

lack of training or lapsed knowledge– it's clear that further CPR training and awareness are a national and global necessity[16]. However, even with training, some remain weary of performing CPR for other reasons. Children and teenagers with CPR training tend to still hesitate to perform resuscitation due to low confidence in their abilities, and they may lack the adequate strength to execute the compression successfully. There tends to be heightened trust in medical professionals and devices than in one's own practices, and due to fear of harming the victim more, bystanders, even with certifiable training, have a general preference to wait for further assistance[16,30]. Rather than perform CPR, the majority of bystanders tend to call for emergency services and seek an on-hand defibrillator, such as an AED. Though AED machines are labeled as "portable" and "easy to use," they usually are only found in gymnasiums, sports arenas, hospitals, or public areas such as shopping malls. CDC found that many cardiovascular-related deaths occur in private locations with about 7 in 10 of the 350,000 cardiac arrests that happen outside hospitals occurring at home[12]. AED machines, such as Philips HeartStart FRx and HeartSine Samaritan PAS 350P, two of the most popular types, are for sale online– however, they, and other brands, cost around $ 1,500, with top models costing more than $ 2000 [85]. The price is a deterrent for low-income (and even middle-to-high-income) families, leading to only a few individuals purchasing these machines for private purposes. As such, AED machines are inaccessible across all income ranges and races, although other factors contribute to lower Sudden Cardiac Arrest survival rates for racial minorities. The cost of the AED machine cannot be easily reduced, mainly due to the high cost of producing the machine's highly accurate microchip. The development of an alternative mechanism may be a plausible solution; however, the current AED is successful in increasing survival chances and reducing long-term neurological damage (Researchers Kroll et al. found there to be a 90% success rate of preserving cognitive function if defibrillation is applied within ten minutes, 50 success rate within 14 minutes following arrest)[43]. Survival rates after ventricular fibrillation cardiac arrest can fall 10% each minute passed without defibrillation[43,53]. Following twelve minutes of cardiac arrest, survival rates, even with CPR, are around 2%[53]. As such, there is no significant flaw with the current design of the AED machine other than it necessitates usage within a pressing time frame. This paper refers to this period in which changes in survival decrease significantly following Sudden Cardiac Arrest as the Critical Period.

Automated External Defibrillators

Automated External Defibrillators, or AED machines, place defibrillation in the hands of laypersons and have been cited as possibly one of the greatest advances in treating Sudden Cardiac Arrest since the development of the CPR protocol. Requiring minimal training, these devices are simple to operate and reliable. As aforementioned, the initial rhythm observed in Sudden Cardiac Arrest is ventricular fibrillation, and though CPR does prolong this arrhythmia, assisting in the preservation of cognitive function, CPR alone is unlikely to convert it to a normal rhythm[53]. The very term "defibrillation" refers to the reversal of the act of fibrillation; it is not a synonym for the term "shock" nor should be referred to as the restoration of a rhythm. The AED incorporates a rhythm analysis system and a shock-advisory system[53]. The microprocessor within the machine analyzes features of the victim's surface ECG signals, such as frequency, amplitude, and wave morphology, along with QRS signals, spontaneous movement, and sources of external problems, such as faulty placement of electrodes or poor electrode contact on the skin[53]. AED machines should only be operated on those who are unresponsive and show no respiratory or circulatory signs (ie, no signs of coughing, movement, or breathing). Dickey W, Dalzell GW, et al. investigated the accuracy of the AED's decision-making capabilities during cardiac arrest; it was found that the accuracy in rhythm analysis was significantly high, and the rare errors were nearly always the result of operators who failed to follow the operating procedures[25]. Failure to follow the audio and visual directions provided by the AED machine can lead to rare instances (<0.1%) where inappropriate electrical counter-shocks have been observed; overall, however, AEDs are often unaffected by artificial signals, the repositioning of the patient, and other variable factors[53]. In general, the machine settings for defibrillators are designed to deliver the lowest energy necessary to terminate ventricular fibrillation; if the current energy levels are too strong, myocardial damage may result[53]. There are two waveform categories of AEDs. Monophasic waveforms deliver energy currents that are 1 polarity (in the direction of current flow) and are then subdivided via the rate at which the current pulse decreases to zero– either gradually or instantaneously[79]. The first shock defibrillation energy for monophasic devices is often around 200J, with the second shock of 200-300J, escalating up to 360J on the third [79]. The second category, biphasic defibrillators, utilizes two energy pulses, with the second having a polarity opposite of the first. Referred to as the impedance-compensating biphasic truncated exponential (BTE) waveform, experimental studies have indicated the superiority of this waveform over monophasic[29, 79]. Gliner and team noted that BTE shocks at low energy currents (115J) were equally effective as 200J monophasic shocks; it was also found that BTE shocks were also superior in terminating ventricular fibrillation with energy as low as 70J[29]. Though further investigation is necessary, the initial findings suggest that BTE waveform defibrillation may provide a safer but equally as effective mode of treatment. Further, no correlation has been observed between body size and energy requirement for AED use.

Adult AED Operation

Having stressed the importance of cardiopulmonary resuscitation training and education, provided for further reference is a brief overview of the common steps towards operating an AED. Following the AED from its box, power on the device via a flip or switch to initiate voice prompts. Cease CPR before the next step. Open and attach the self-adhesive electrode pads to the victim's skin; place the first pad on the upper-right sternal border (located on the right side of the patient's sternum, below the collar bone). The second pad is to be placed on the left mid-axillary line, laterally placed clear of breast tissue (on the patient's left side, horizontally put under the breast). If the voice prompts to "check electrodes," press firmly on each pad. If the error continues, remove the original pads and shave the chest area of the pads before attaching another set. Next, ensure that bystanders are not touching the victim. Some AED models have an "Analyze" button, while others initiate rhythm analysis automatically. Analysis requires around five-fifteen seconds. If ventricular fibrillation is present, the voice operator will sound a short alarm to indicate that shock is required. Again, ensure nobody is touching the victim prior to pressing the "Shock" button. Do not restart CPR following the first shock. Press "An-

alyze" again if it is not automatic, and if ventricular fibrillation is present, the AED will note it, and the sequence will repeat for a second and if necessary, a third shock. Following the third shock, perform CPR again for one minute.

Pediatric AED Operation

The use of AEDs on infants and children <8 years of age is not recommended, and victims should weigh >25kg (55 lbs)[29,79]. Rather, pediatric CPR should be performed. Patients in water or with pacemakers/ICDs are further circumstances that may require responders to modify the protocol[53]. Cardiac arrest in children is uncommon compared to adults, with around 50% of pediatric cases occurring in infants younger than age one[53]. In general, the sensitivity and specificity of the AED algorithm necessitate further research, as the current experience of defibrillation and children is scarce. Determining the energy pulse levels for children is posing the primary limitation towards determining defibrillation that is only safe but effective in young children[25]. In situations concerning water, victims of Sudden Cardiac Arrest should be removed from freestanding water and dried prior to AED operation. As water is a natural and effective conductor of electricity, the bystanders may experience minor burns if within a water pathway. Further, water between the electrode pad and the epidermis can decrease the effectiveness of the delivered shock. Patients with implanted pacemakers/ICDs often have a history of malignant arrhythmias[24]. ICDs or implanted cardioverter defibrillators function similarly to a pacemaker (an implanted device that generates electrical signals to control heart rate rhythm) but are capable of sending low electrical shocks directly to the myocardium to reset an arrhythmic heart rate back to normal[53]. Normally, implanted devices are visible as hard lumps beneath the skin of the left side of the victim's abdomen or chest; when an AED electrode is placed directly over an ICD or pacemaker, the effectiveness of defibrillation is reduced. As such, one should place the electrode pad an inch away from the device to avoid interference and improve effectiveness. Other than the aforementioned circumstances (water, ICDs/Pacemakers, and infants) there are no other requirements for defibrillation. As such, the AED is often universally applicable and thereby, often successful if operated immediately following arrest.

Significant Concerns and Proposed Solutions

A significant flaw with the AED machine is that it must be used during the critical period after cardiac arrest. The AED device, on its own, is not a cause for concern. With its low error rates, it can significantly increase the survival probability following cardiac arrest[25]. That being said, there is always room for improvement in the device's design to enhance its effectiveness further. The primary concern contributing to cardiac arrest as a leading cause of death worldwide is the difficulties bystanders face in accessing an AED. The difficulty in accessing an AED device efficiently is a constraint that prevents achieving better survival outcomes.

The Medical Emergency Box Proposal

70-80% of Sudden Cardiac Arrests occur in private locations like homes[12]. The lack of access in private locations makes it not merely hard, but nearly impossible for individuals to access AEDs. A proposed solution is to work with city governments to implement AED accessibility policies. Specifically, I offer a model similar to that of "Little Libraries." Though "Little Libraries" are public book exchanges and have nothing to do with the medical field, this proposal makes use of the concept that this system utilizes– making commodities easy to access within a neighborhood. Medical Emergency Boxes, or MEB for short, would bring Automated External Defibrillators to neighborhoods and rural areas as a public good **(Figure 6).** Designed out of sustainable and weather-resistant material, the exterior would be composed of Polycarbonate, Cork, and Recycled Steel, with the interior coated with long-lasting Plant-based Polyurethane Rigid Foam (PbPRF). With a high R-Value, PbPRF acts as an effective fire-resistant insulator and protects against vapor, mold, and pests. A functioning AED will be placed within each MEB, which will be accessible to anybody who requires it. In the bottom compartment of the box would be a medical supply vending machine, though one would not have to pay to have access **(Figure 6).** Here CPR disease-prevention masks, additional razors for the removal of excess chest hair, medical scissors to cut through clothing, and other medical aid appliances that may be of use to the AED operator can be found. Though some AEDs are equipped with such additional supplies at times, they are not always reliable; the MEB takes this into account and takes additional precautionary measures. Much like the Blue Light Alarm System in place at many college campuses within the United States, the MEB would be equipped to call emergency services and would do so automatically when the box is open **(Figure 6)**. There would also be a flashing light located on the top of the box, which would turn on when the box is open. These two apparatuses also act as a deterrent against crime or vandalism on the MEB. A video surveillance camera may be of use as well. Policy-wise, a fine for inappropriately using or damaging such a box would also be implemented as a safety measure. On the side of the MEB could be instructional pamphlets or with advancing technologies, a screen that can provide an individual with steps for procedures such as how to perform CPR, treat a wound, or work an Epipen. This may also be a QR code that those passing the street would be able to scan for future use. In doing so, CPR and first-aid knowledge become more publicly accessible as well. To further increase accessibility, the instructions should also be translatable into a variety of different languages. Though the initial proposal was for the MEB to merely supply one with the contents necessary only for cardiac arrest it is possible, if the government or city budget allows, for the MEB to also contain other medical supplies for other emergencies such Naloxone, which, if administered in time, can reverse an Opioid overdose or extra Epipens for severe allergic reactions. The logistics for a suburban neighborhood, which is also applicable to a rough outline for rural and metro areas, are as follows.

On average, it is often the case that eight American street blocks equate to a distance of a mile, or 5280 feet. According to a 2019 study, the majority of adult individuals can expect to walk a mile in 15 to 22 minutes at the reported average walking pace of around 2.5 to 4mph [58]. As such, at a jogging speed of say between 4mph-6mph, a healthy individual should be able to reach the end of a four-block distance (half a mile) in a time frame between three-to-six minutes, depending on the pace[58]. As such, placing an MEB every four blocks within an area would be sufficient in increasing accessibility to medical devices, such as AED machines. To further aid in finding MEBs, arrows drawn on street curbs (much like painted house numbers) can point toward the location of the nearest one. According



Figure 6 Hand-Drawn prototype design of the Medical Emergency Box (MEB). The exterior would be composed of Polycarbonate, Cork, and Recycled Steel, with the interior coated with long-lasting Plant-based Polyurethane Rigid Foam (PbPRF).

to the United States Department of Agriculture, a rural area consists of open countryside, often with population densities of less than 2,500 people or less than 500 people per square mile[18]. As such, the placement of MEB would need to change appropriately. As such, I propose placing an MEB at the edge of each square mile. If this is not an applicable measure, I suggest following the National Fire Hydrant Guidelines for a specific area to determine the placement of MEBs. In general, these Medical Emergency Boxes should be placed in a recognizable location, easily accessible via car, bike, or jog. The placement of an MEB is of great importance– not knowing the location of an AED in time of a crisis contributes to lower survival rates. MEB would act as a system that would help equalize the medical field, providing AED access to all, regardless of socioeconomic standing. Further, in making AEDs a public good located not merely in stadiums, airports, or shopping malls, but within the family bounders of one's local residence, it further combats the problem of lengthy travel times to emergency services.

The Find My AED App Proposal

There is an overall lack of knowledge as to the location of the nearest AED machine even for those residing in neighborhoods, apartment complexes, or metropolitan areas. It is not often common knowledge where an AED is located– even within parks, stadiums, office buildings, and other public places, locating the red box can prove difficult. For metropolitan areas specifically, following the aforementioned guidelines for MEB placements would not be cost-effective, as office buildings and other public facilities would often already have an AED on its premise. However, most do not know where the nearest AED is located (or the specific location of the device within a crowded or large area). An app or website that could bring up the locations of the nearest AED with respect to victims of cardiac arrest, however, could help resolve this problem. I propose developing a program in which the bystander can type their present address or utilize the phone's automatic location-placing ability to then highlight the nearest AED machines with approximated times of arrival (walking distance). When the bystander makes a judgment as to which device is closest to reach, the app would provide detailed instructions as to how to get there. There are currently two apps that perform this function. Furthering awareness of these apps is necessary. Rather than it merely being an App one has to download and actively search, it should be a program already a part of the phone, much like how "Find My" or "Maps" are.

Apple Watch Proposal

There is another pressing concern that neither MEB nor an AED-Finder Program can solve: also applicable to CPR, the AED device requires there to be a spectator near the victim at the time of the cardiac arrest. If one were to suffer from cardiac arrest by their lonesome (ie, in the bathroom, living alone, in the car, or in other isolated environments), access to emergency medical services or devices often occurs too late. When alone, cardiac arrest survival rates plummet to 10% within the first few minutes, compared to the 90% survival rate when a bystander and AED are on hand[12,63,76]. Perhaps developing an AED-similar device that one can possibly apply to themselves during the chest-pain phase would act as a possible solution (though this does not account for the economic inaccessibility of the AED's microchip, which would be used in a self-applied AED). Another solution is to utilize already popularized and available algorithms, such as that in the Apple Watch ECG and Heart Rate Monitoring applications[39]. Currently, any Apple Watch past Series 4 can record cardiac rhythm and trends, alert wearers of dangerously high or low heart rates, and detect atrial fibrillation, a form of irregular rhythm[39]. If Apple or similar companies were to enhance this software so that it could detect any ventricular and tachycardia fibrillation, not only atrial, or could alert 911 when the heart rate suddenly stops, emergency services may be able to reach the lone patient within the Critical Period. At the very least, the software, when detecting arrhythmia or similar concerning cardiac signals, should immediately alert a family member (or another emergency contact) of the situation and send them the location of the individual automatically. This way, the software circumvents the concern of accidentally and unnecessarily altering medical professionals but still alerts others of the possibility of a friend or family member undergoing cardiac arrest. This would be especially useful for seniors who may live alone and away from their adult children. Even if the contact is far from the individual at the time of the notification, it still makes them aware of the situation, and they can act by calling emergency services to the location or by sending somebody to check on them.

Conclusion

This work hoped to bring forth a concern observed in the medical field regarding the inaccessibility of Automated External Defibrillators (AED). Sudden Cardiac Arrest is the leading cause of death globally, and the current treatment for Sudden Cardiac Arrest– specifically the use of AED machines– is an insufficient solution as there is a lack of equal access to such a device due to

socioeconomic factors[36,81]. Not only are AEDs financially inaccessible to lower-to-middle income families, but they are typically found in public areas, such as shopping malls or schools; many cardiovascular-related deaths occur in private locations, away from an easily accessible AED machine. Further, within the first few minutes of cardiac arrest, the victim requires immediate attention to increase the chance of survival– a period referred to as the critical period. One of the primary concerns of the modern defibrillator is the correlation between the device's success in preventing on-site death and the time period between victim collapse and AED use. An AED must be used within a matter of minutes for optimal success, and the overall inaccessibility to these devices makes it near impossible for many, especially those living in low-income areas, which are often suburban areas further from hospital sites, to have access to the machine during that critical time frame. In general, efforts to improve cardiopulmonary resuscitation bystander engagement and quality of treatment in medical institutions are necessary to combat the low Sudden Cardiac Arrest survival rates. Though this paper was written from an American perspective, which affects the author understanding regarding the structure of the medical system and the organization of metropolitan, urban, and rural areas, the suggested proposals welcome further refinement and necessary adjustments to apply to other countries as well. Methods that push for improved accessibility to AED machines– AED-locating apps, public neighborhood Medical Emergency Box systems, utilizing already implements software in devices such as Apple Watches– may provide a short-term solution to combating this public health concern.

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